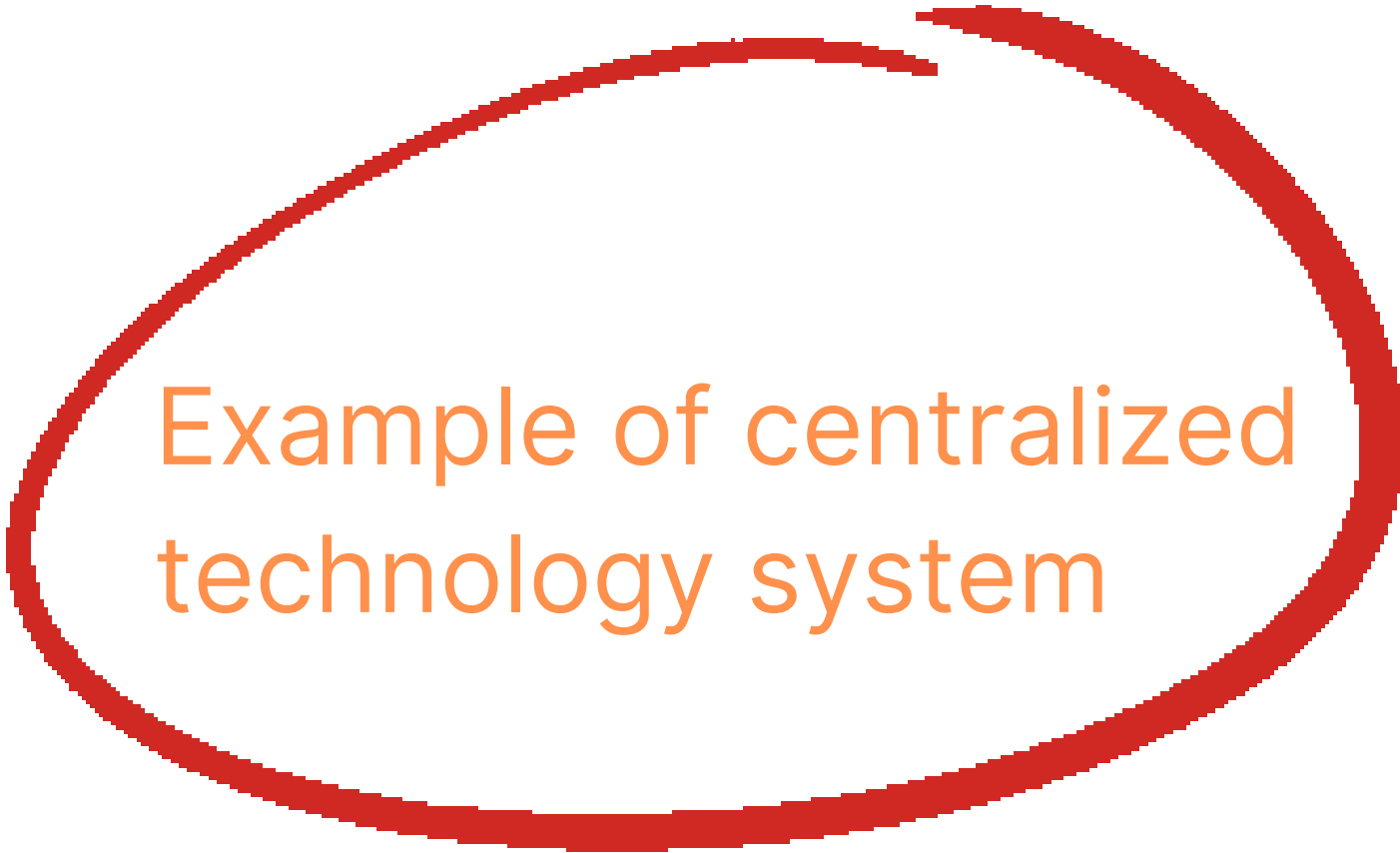




BLOCKCHAIN TECHNOLOGY

PROBLEM OF CENTRALIZED TECHNOLOGY

- 01 Security
- 02 Single point of failure
- 03 Data privacy
- 04 Transparency



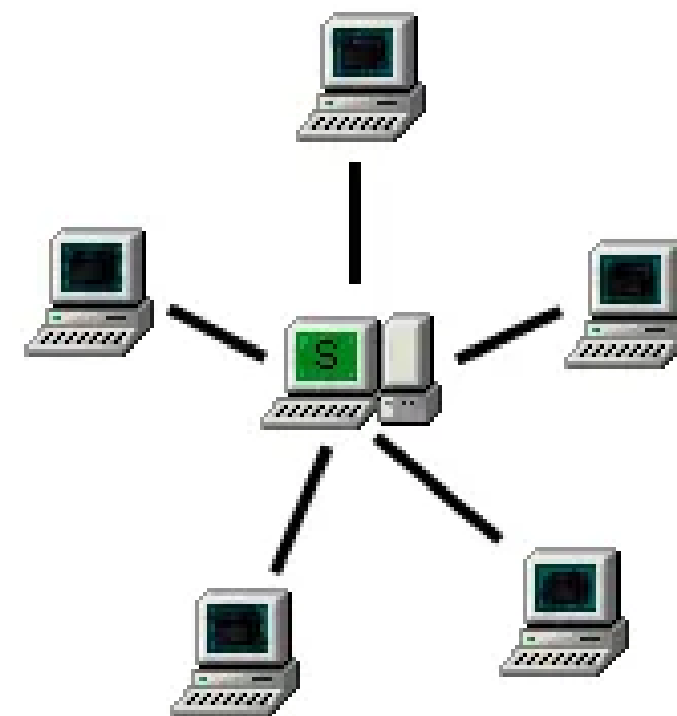
Example of centralized
technology system

Technologies behind blockchain

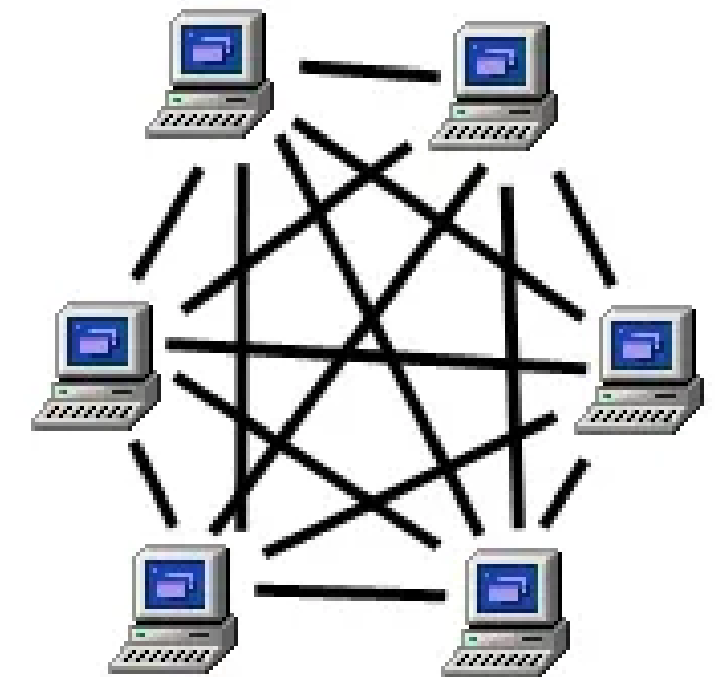
- 01 Peer-to-peer network
- 02 Distributed ledger technology
- 03 Cryptography

Example of peer-to-peer
market

Server Based Network



Peer to Peer Network



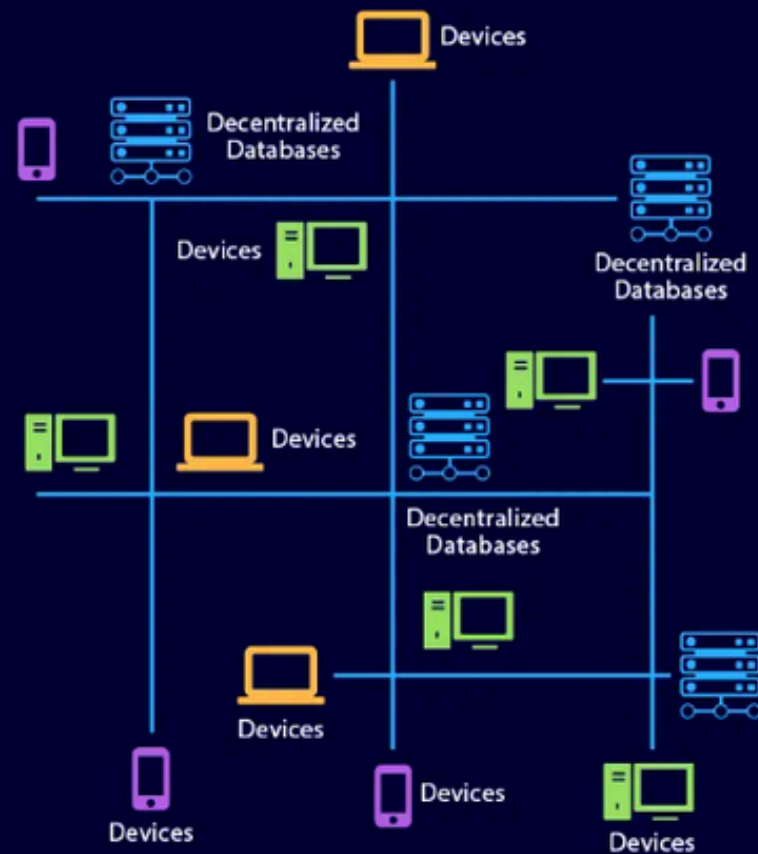
Centralized -- Decentralized -- Distributed

Centralized vs Decentralized Internet

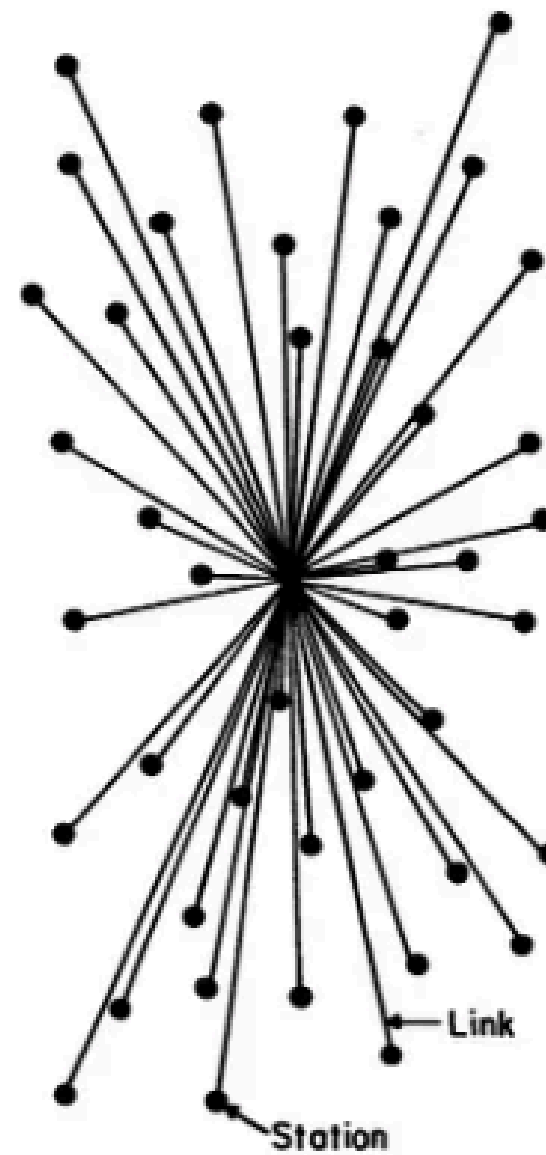
BEFORE



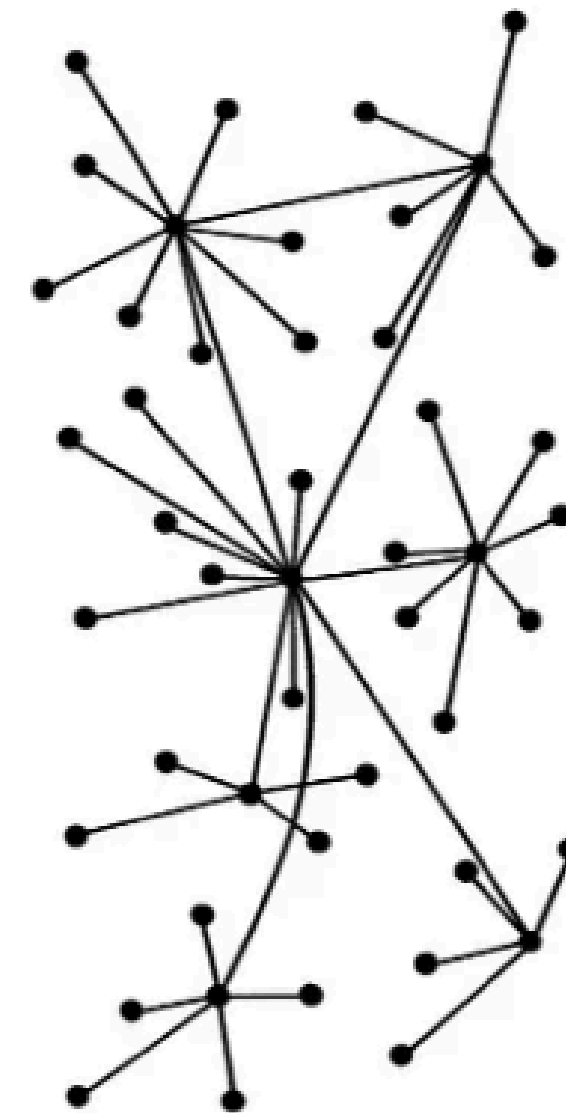
AFTER



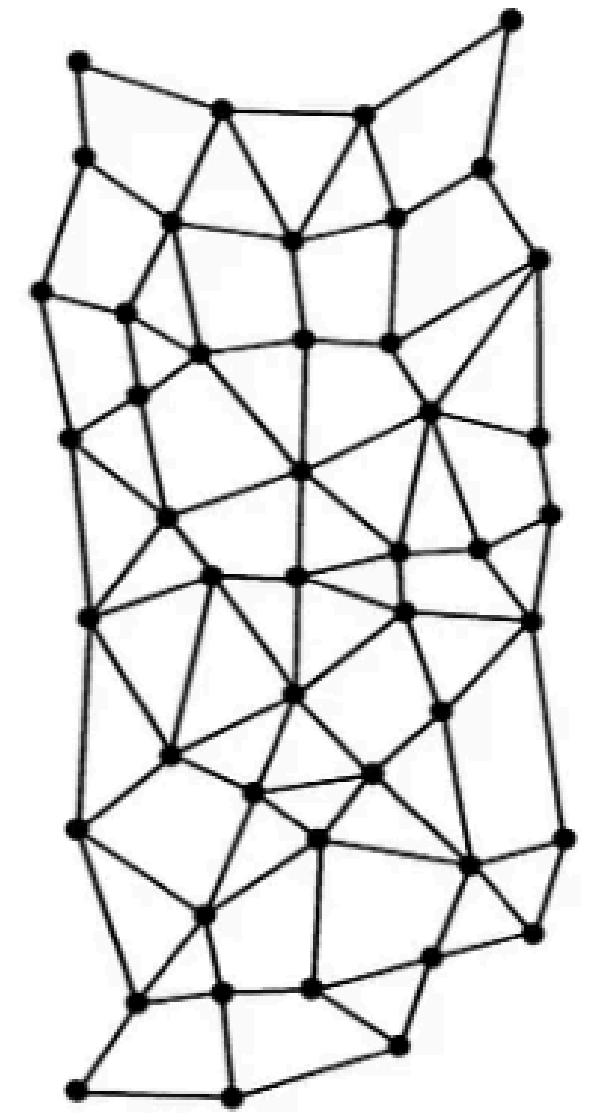
 **101 Blockchains**
Created by 101blockchains.com



CENTRALIZED
(A)



DECENTRALIZED
(B)



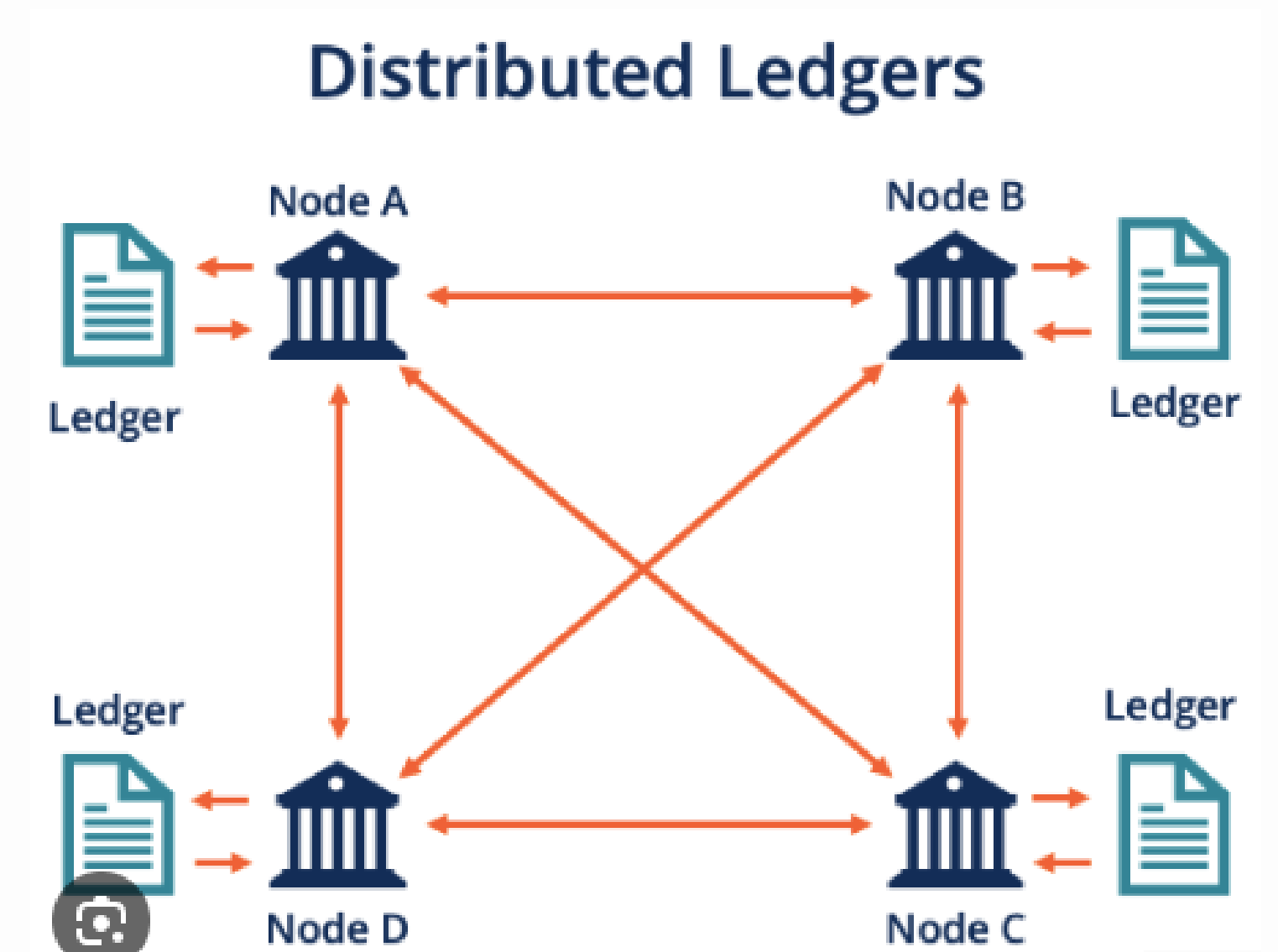
DISTRIBUTED
(C)

Distributed Ledger Technology

Distributed Ledger Technology (DLT) เป็นการบันทึกข้อมูลและธุรกรรมแบบออนไลน์โดยข้อมูลจะถูกจัดเก็บไว้หลาย ๆ ที่ในเวลาเดียวกัน

คุณลักษณะของ DLT

- 01 การกระจายข้อมูล (decentralization)
- 02 ฉันทามติ (consensus)
- 03 ความโปร่งใส (transparency)
- 04 ความปลอดภัย (security)



BLOCKCHAIN

Blockchain (บล็อกเชน) คือ ระบบการเก็บข้อมูลแบบกระจาย (แบบหนึ่งของ DLT) ที่มีความน่าเชื่อถือ โปร่งใส และไม่ต้องอาศัยคนกลาง

What is Blockchain?

Blockchain is a decentralized and distributed digital ledger.

It is a chain of blocks that store transactional records.



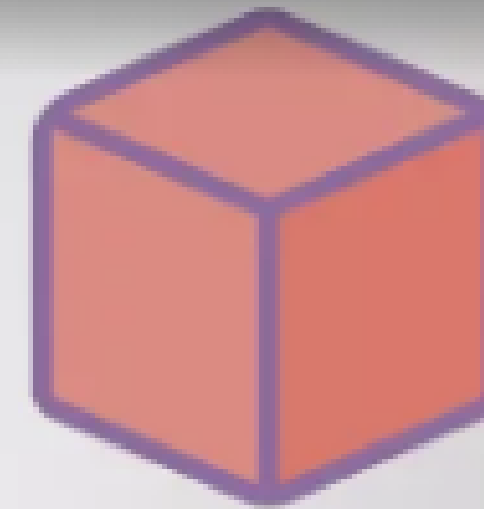
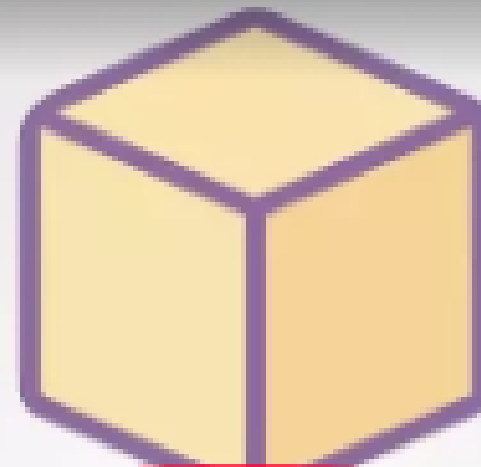
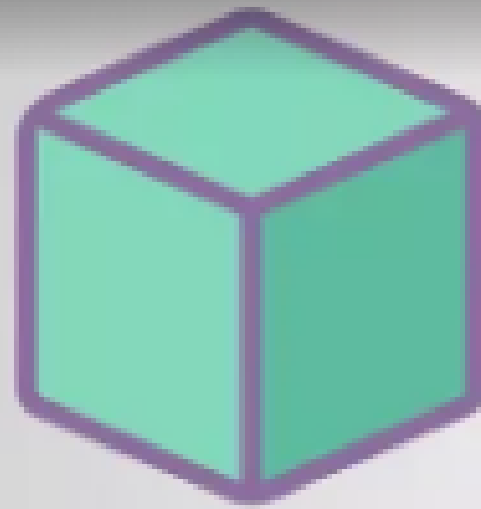
It is transparent, secure, and immutable.

It eliminates the need for intermediaries.



How does a blockchain work - Simply Explained

Simply Explained

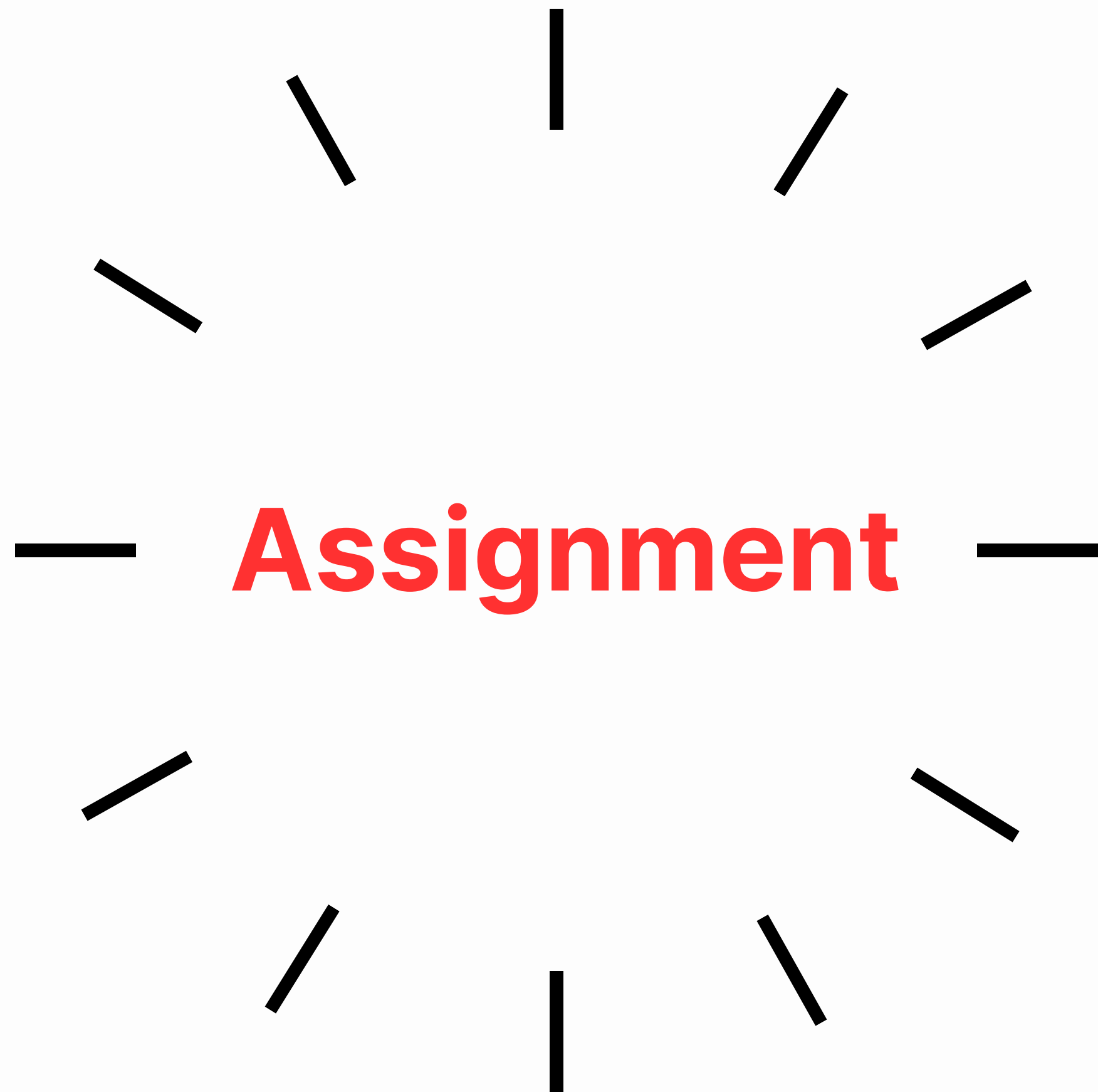


Blockchain

— *Simply explained* —



Watch on  YouTube



Cryptography

Cryptography คือ เทคโนโลยีที่เกี่ยวข้องกับการป้องกันข้อมูลและความเป็นส่วนตัวโดยใช้เทคนิคการเข้ารหัสข้อมูล

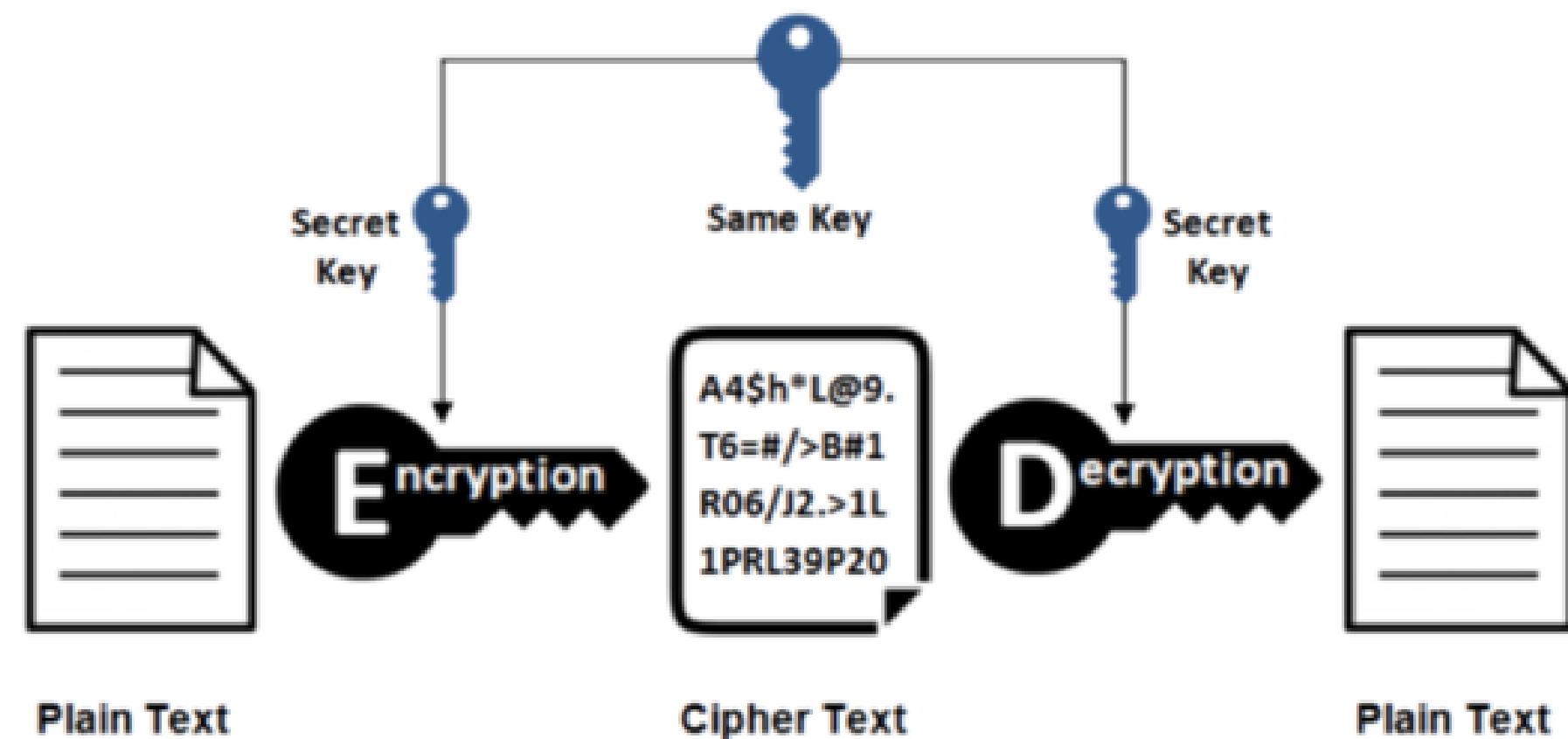
Cryptographic techniques used in blockchain

1) Public-key cryptography

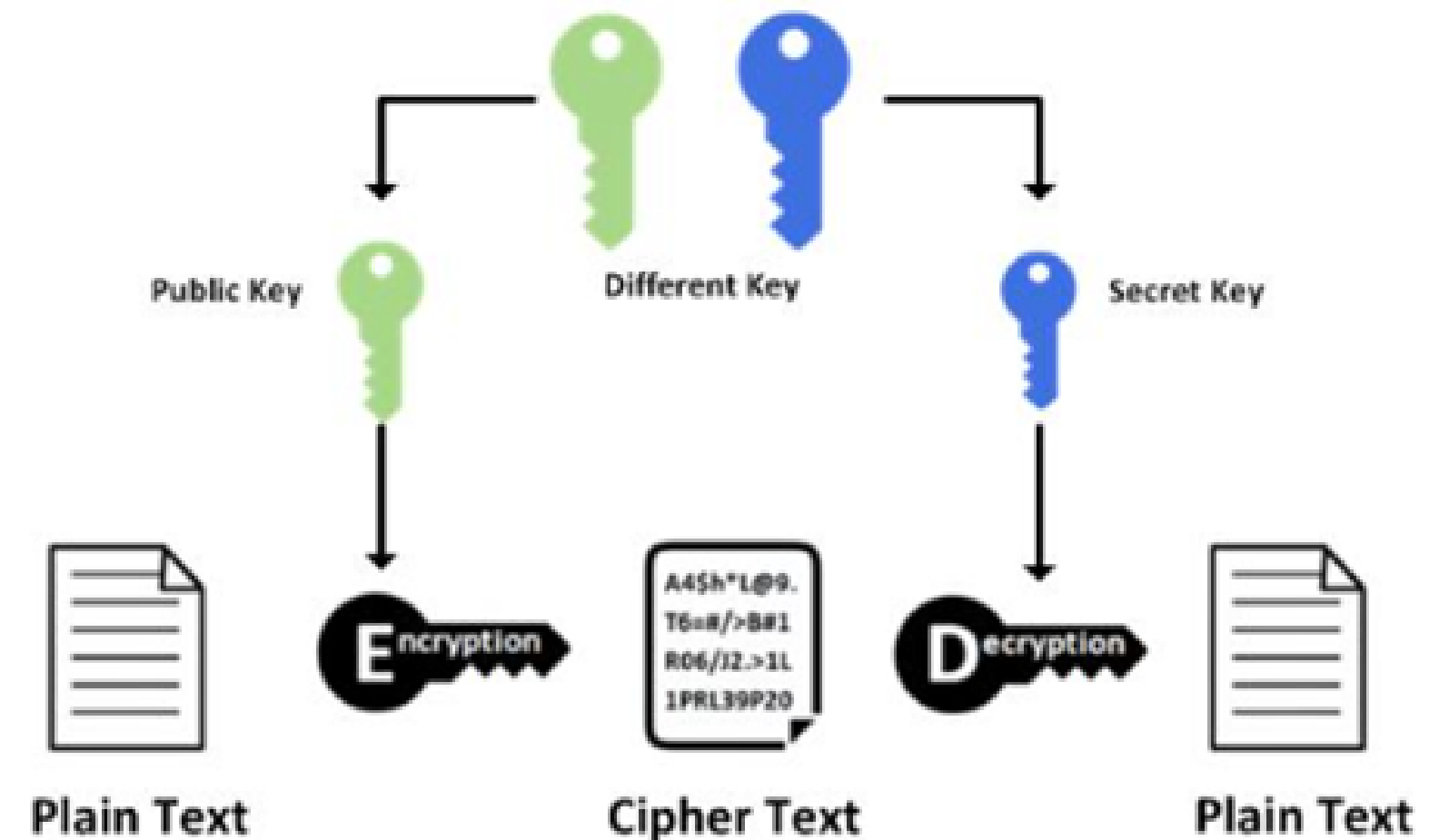
Public-key cryptography คือ การเข้ารหัสแบบกุญแจสาธารณะ หรือบางครั้งเรียกว่า asymmetric cryptography ซึ่งจะตรงข้ามกับ symmetric cryptography

Symmetric vs Asymmetric encryption

Symmetric Encryption



Asymmetric Encryption



Cryptographic techniques used in blockchain (cont.)

2) Hash function

อัลกอริทึมหรือฟังก์ชันที่รับข้อมูลอะไรก็ตามในรูปแบบใดก็ตามและสร้างเอาต์พุต (output) เป็นสตริงความยาวคงที่ โดยฟังก์ชันแฮชจะทำงานโดยรับข้อมูลเข้าและผ่านกระบวนการทางคณิตศาสตร์เพื่อสร้างแฮชของข้อมูลนั้น ๆ

มีคุณลักษณะดังนี้

- ไม่ว่า input จะมีขนาดเท่าไร output จะมีขนาดเท่ากัน
- ไม่สามารถย้อนกลับข้อมูลเดิมจากข้อมูลที่ถูกเข้ารหัส (one-way function)

Hashing Algorithm in Blockchain



<https://medium.com/dcen/hashing-algorithm-in-blockchain-367f09e043cf>

<https://emn178.github.io/online-tools/sha256.html>





**Continue get
token**

Workshop

Get free token from

1) Sepolia faucet

Workshop

!!! Make sure you have 2 Metamask wallets

Connect and get free token from testnet

1) Sepolia faucet (SepoliaETH)

[Link to ChainLink documentation](#)

Workshop

!!! Make sure you have 2 Metamask wallets

2) OpenZeppelin

3) Remix IDE

4) Deploy contract to Sepolia Testnet

5) Check block explorer

Transaction Hash: 0xa459e1f8d2b2df143e59087986bfa04b03f48de8a6b4819f72d5f988d774e463

Status: Success

Block: 11127678 43910 Block Confirmations

Timestamp: 6 days ago | Jun-24-2026 04:59:24 AM +UTC

From: 0x317fc4b53d9fa6dE4407d16b0014F4d866a67fEe

To: 0x3420a3f31DF2b3E5F473e78632d3Dc3f19388A3B

Value: 0.01 ETH

Transaction Fee: 0.000054357540699 ETH

Gas Price: 2.588454319 Gwei (0.000000002588454319 ETH)

Gas Limit & Usage by Txn: 21,000 | 21,000 (100%)

Gas Fees: Base: 1.088454319 Gwei | Max: 2.99095871 Gwei | Max Priority: 1.5 Gwei

Burnt & Txn Savings Fees: Burnt: 0.000022857540699 ETH (\$0.00) Txn Savings: 0.000008452592211 ETH (\$0.00)

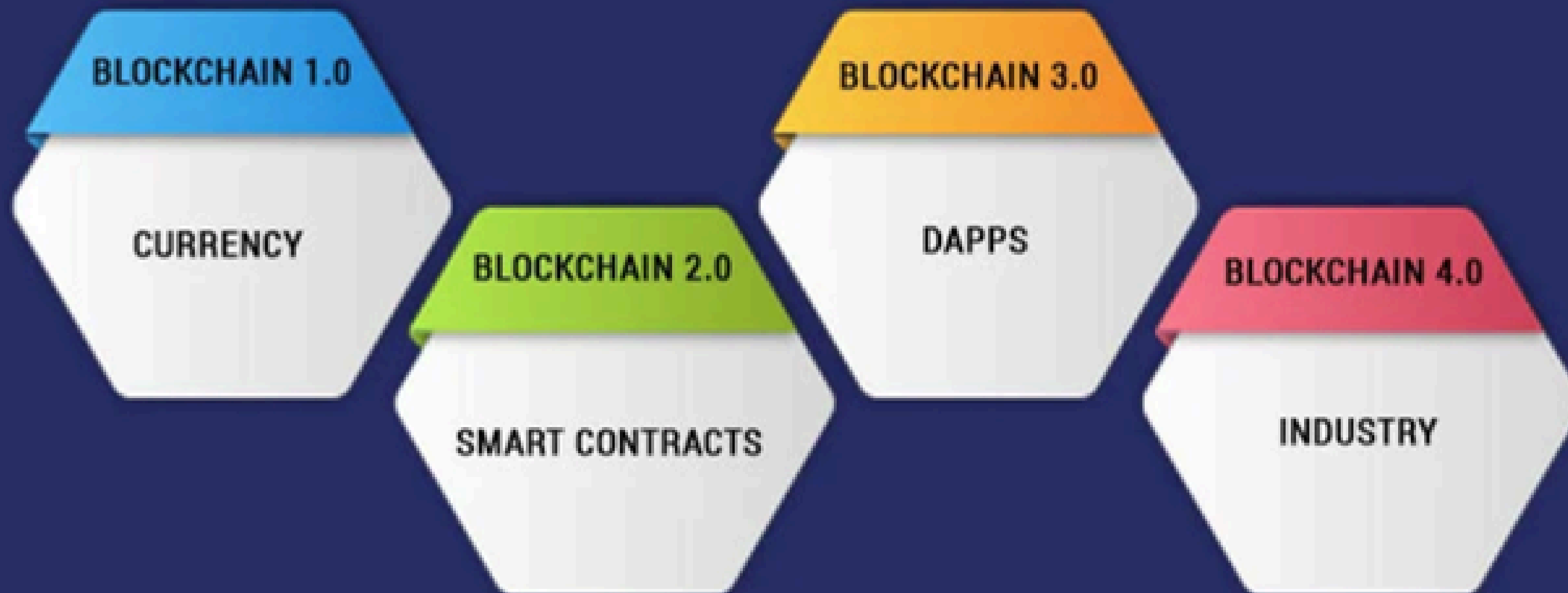
Other Attributes: Txn Type: 2 (EIP-1559) Nonce: 838 Position In Block: 28

Input Data: 0x

Blockchain

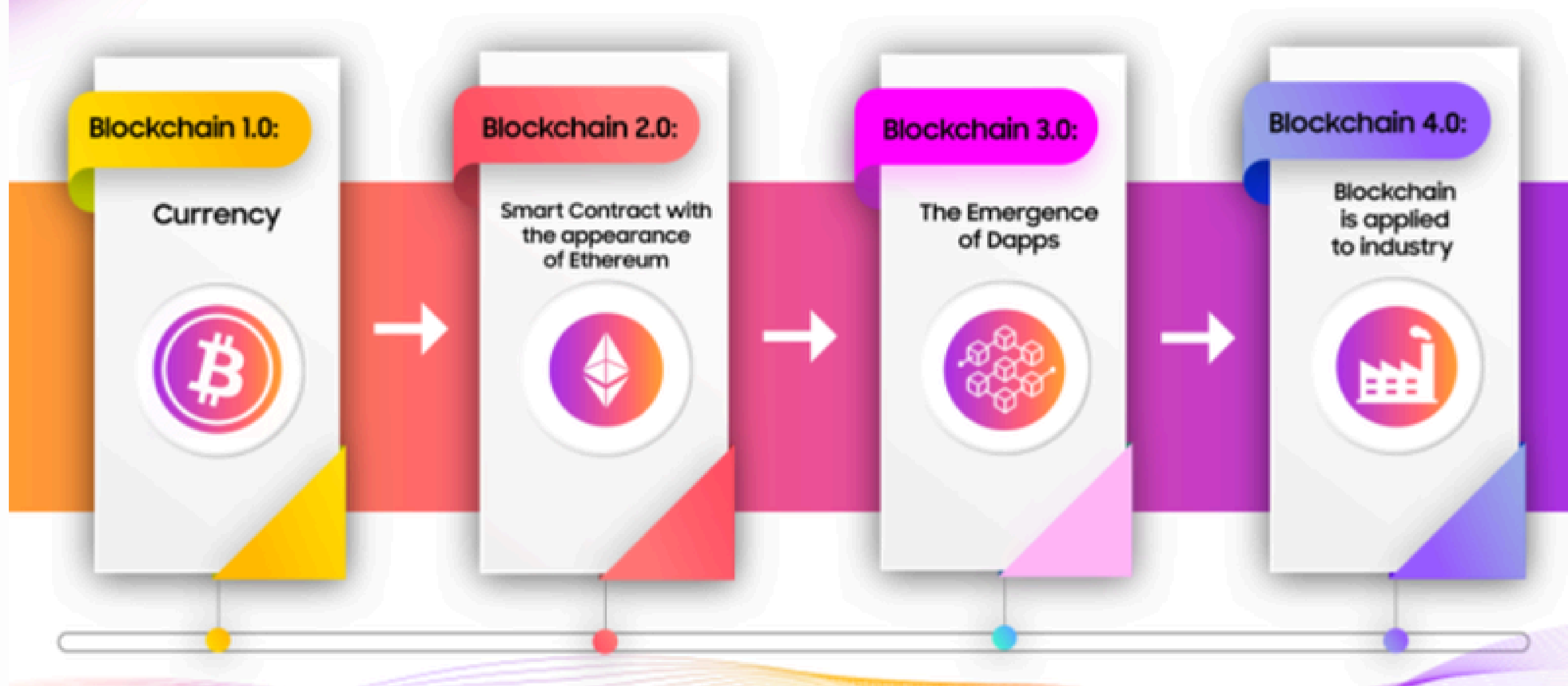
ข้อมูลที่ได้รับการจัดเก็บใน blockchain จะไม่สามารถเปลี่ยนแปลงหรือแก้ไขได้ (immutability) จึงทำให้เทคโนโลยีนี้เป็นที่ยอมรับว่าเป็นเทคโนโลยีสำหรับการจัดเก็บข้อมูลที่มีความปลอดภัย (security) โปร่งใส (transparency) และมีความน่าเชื่อถือสูง (reliability)

BLOCKCHAIN HISTORY AND EVOLUTION



<https://www.ifourtechnolab.com/blog/blockchain-history-and-evolution>

Blockchain Evolution



Blockchain.com

Home

Prices

Charts

NFTs

DeFi

Academy

News

Developers

Wallet

Exchange

Bitcoin

Ethereum

Bitcoin Cash

Search Blockchain, Transactions, Addresses and Blocks

Tether/USD 1.00

▲0.00%

XRP/USD 0.79

▼4.35%

Binance Coin/USD 243.93

▲1.05%

Wise Token/USD 47.45

▲0.05%



Bitcoin BTC

Bitcoin (BTC) is a decentralized currency that eliminates the need for central authorities such as banks or government to-peer internet network to confirm transactions directly between users.

Price History

\$29,887.19

+ 11:10

Vol 26 BTC

1D

1W

1M

1Y

MAX

USD



Blockchain.com

Home

Prices

Charts

NFTs

DeFi

Academy

News

Developers

Wallet

Exchange

Bitcoin

Ethereum

Bitcoin Cash

Search Blockchain, Transactions, Addresses and Blocks

OKB/USD 42.91 -0.04%

Dogecoin/USD 0.07 +2.43%

Stellar/USD 0.17 -3.01%

Chainlink/USD 8.31 +19.96%

Aptos/USD 7.49 -0.68%

Polygon/USD 0.77 +0.76%

TRON/USD 0.10 +0.01%

Blockchain

#799610

#799609

#799608

#799607

#799606

#799605

#799604

#799603

#799602

#799601

#799600

#799599

#799598

Market Info

Market Cap ⓘ
\$580,898,858,987

Diluted Market Cap ⓘ
\$627,674,249,309

Vol 24h ⓘ
\$587,736,810

Vol / Market Cap ⓘ
0.0010

24h Change ⓘ
-0.26%

1h Change ⓘ
-0.05%

24h High ⓘ
\$30,419.86

24h Low ⓘ
\$29,569.02

Latest Block
799,606

Block Reward
6.25

Hashes Per Second
-

Circulating Supply ⓘ
19,435,043

Max Supply ⓘ
21,000,000

Consensus ⓘ
Proof of Work

Algorithm ⓘ
SHA-256

Genesis Block Date ⓘ
2009-01-03

Bitcoin Block 0

Mined on January 04, 2009 01:15:05 • [All Blocks](#)

- Satoshi
- Notable Block

Coinbase Message • The Times 03/Jan/2009 Chancellor on brink of second bailout for banks

Bitcoin Genesis

On January 3rd 2009, the Bitcoin network was created when Satoshi Nakamoto (the project's mysterious creator) mined the "Genesis" block. The 50 bitcoin coinbase reward is unredeemable, as it was omitted from the transaction database. This means any attempt to spend it would be rejected by the network. Whether this was intentional or not still remains unknown.

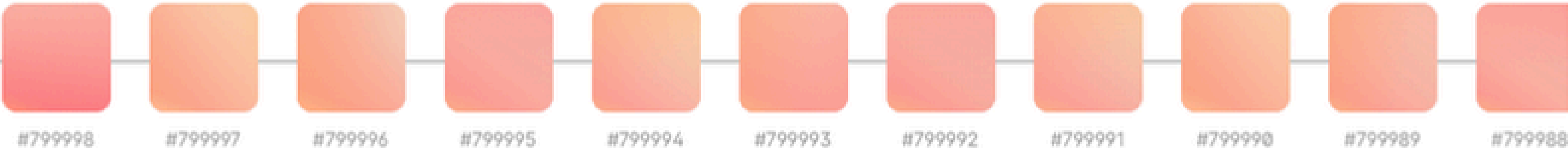
A total of 0.00 BTC (\$0.00) were sent in the block with the average transaction being 0.0000 BTC (\$0.00). Satoshi earned a total reward of 50.00 BTC \$0.00. The reward consisted of a base reward of 50.00 BTC \$0.00 with an additional 0.0000 BTC (\$0.00) reward paid as fees of the 1 transactions which were included in the block.

Details

Hash	00000-ce26f 	Depth	799,998
Capacity	0.03%	Size	285
Distance	14y 6m 19d 8h 49m 45s	Version	0x1
BTC	0.0000	Merkle Root	4a-3b 
Value	\$0.00	Difficulty	1.00
Value Today	\$0.00	Nonce	2,083,236,893
Average Value	0.000000000000 BTC	Bits	486,604,799
Median Value	50.0000000000 BTC	Weight	1,140 WU
Input Value	0.00 BTC	Minted	50.00 BTC
Output Value	50.00 BTC	Reward	50.0000000000 BTC
Transactions	1	Mined on	Jan 04, 2009, 1:15:05 AM
Witness Tx's	0	Height	0
Inputs	1	Confirmations	799,998
Outputs	1	Fee Range	0-0 sat/vByte
Fees	0.0000000000 BTC	Average Fee	0.0000000000
Fees Kb	0.00000000 BTC	Median Fee	0.0000000000
Fees kWU	0.00000000 BTC	Miner	Satoshi

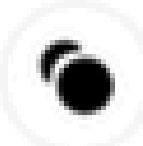
Blockchain

→



Transactions





ID: **4a5e-a33b**

1/04/2009, 01:15:05

From Block Reward

To **1A1z-vfNa**

50.00000000 BTC • \$1,488,830

Fee 0 Sats • \$0.00

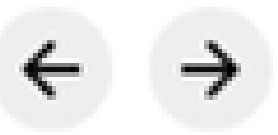
^

From

 1 Block Reward
0.00 BTC • \$0.00

To

1 **1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa**  
50.00000000 BTC • \$1,488,830



Bitcoin Block 799,998

Mined on July 24, 2023 10:03:53 • [All Blocks](#)

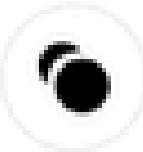
M4

Coinbase Message •     Q:  t     

A total of 13,914.86 BTC (\$414,652,759) were sent in the block with the average transaction being 4.4800 BTC (\$133,500). M4 earned a total reward of 6.25 BTC \$186,245. The reward consisted of a base reward of 6.25 BTC \$186,245 with an additional 0.2036 BTC (\$6,067.13) reward paid as fees of the 3,106 transactions which were included in the block.

Details

Hash	00000-2b5b4 	Depth	1
Capacity	158.54%	Size	1,662,401
Distance	14y 6m 19d 8h 54m 2s	Version	0x310d0000
BTC	13,914.8630	Merkle Root	39-96 
Value	\$414,652,759	Difficulty	53,911,173,001,054.59
Value Today	\$414,172,000	Nonce	1,394,660,396
Average Value	4.4799945213 BTC	Bits	386,218,132
Median Value	0.00165310 BTC	Weight	3,992,687 WU
Input Value	13,915.07 BTC	Minted	6.25 BTC
Output Value	13,921.32 BTC	Reward	6.45357181 BTC
Transactions	3,106	Mined on	Jul 24, 2023, 10:03:53 AM
Witness Tx's	2,826	Height	799,998
Inputs	6,911	Confirmations	1
Outputs	8,642	Fee Range	0-461 sat/vByte
Fees	0.20357181 BTC	Average Fee	0.00006554
Fees Kb	0.0001225 BTC	Median Fee	0.00002047
Fees kWU	0.0000510 BTC	Miner	M4

	0 ID: f31c-bb92  7/24/2023, 10:03:53	From Block Reward To 2 Outputs	6.45357181 BTC • \$192,311 Fee 0 Sats • \$0.00	▼
TX	1 ID: d4d3-c640  7/24/2023, 09:52:07	From bc1p-ytjr  To 2 Outputs	0.00151686 BTC • \$45.20 Fee 94.5K Sats • \$28.17	▼
TX	2 ID: 10b3-90fc  7/24/2023, 09:52:11	From bc1p-uq8m  To 2 Outputs	0.00015208 BTC • \$4.53 Fee 105.8K Sats • \$31.53	▼
TX	3 ID: a1c9-1a3b  7/24/2023, 09:51:18	From bc1q-vzej  To 2 Outputs	0.13624316 BTC • \$4,059.95 Fee 90.0K Sats • \$26.82	▼
TX	4 ID: 796b-f69d  7/24/2023, 09:47:33	From bc1q-vzej  To 2 Outputs	0.06896315 BTC • \$2,055.05 Fee 60.0K Sats • \$17.88	▼
TX	5 ID: ede1-3fd7  7/24/2023, 09:54:50	From bc1q-pnuy  To bc1q-6yej 	0.06845185 BTC • \$2,039.82 Fee 33.0K Sats • \$9.83	▼
TX	6 ID: d61c-35dc  7/24/2023, 09:56:58	From bc1q-j0h2  To 2 Outputs	1.99957700 BTC • \$59,585.93 Fee 42.3K Sats • \$12.61	▼
TX	7 ID: f854-586e  7/24/2023, 09:54:19	From bc1q-mnjq  To 2 Outputs	552.24476474 BTC • \$16,456,490 Fee 40.3K Sats • \$12.01	▼

Transactions

Last

First

Value

Value

Fee

Fee

0 ID: f31c-bb92

7/24/2023, 10:03:53

From Block Reward
To 2 Outputs

6.45357181 BTC • \$192,311
Fee 0 Sats • \$0.00

From	To
1 Block Reward 0.00 BTC • \$0.00	1 3KZDwmJHB6QJ13QPXHqW7SS3yTESFPZoxb 6.45357181 BTC • \$192,311 2 Unknown 0.00000000 BTC • \$0.00

TX

1 ID: d4d3-c640

7/24/2023, 09:52:07

From bc1p-ytjr
To 2 Outputs

0.00151686 BTC • \$45.20
Fee 94.5K Sats • \$28.17

From	To
1 bc1pk48s0mfwca8036tyq7vsdxwc54vyr54224hfn4c3grz50p2... 0.00246230 BTC • \$73.37	1 bc1pv6ar4n5ex7g0sf8h667xmlv0g6k8d5j2zrlwgz343scnxkxpl0... 0.00121000 BTC • \$36.06 2 bc1pk48s0mfwca8036tyq7vsdxwc54vyr54224hfn4c3grz50p2sh... 0.00030686 BTC • \$9.14

TX

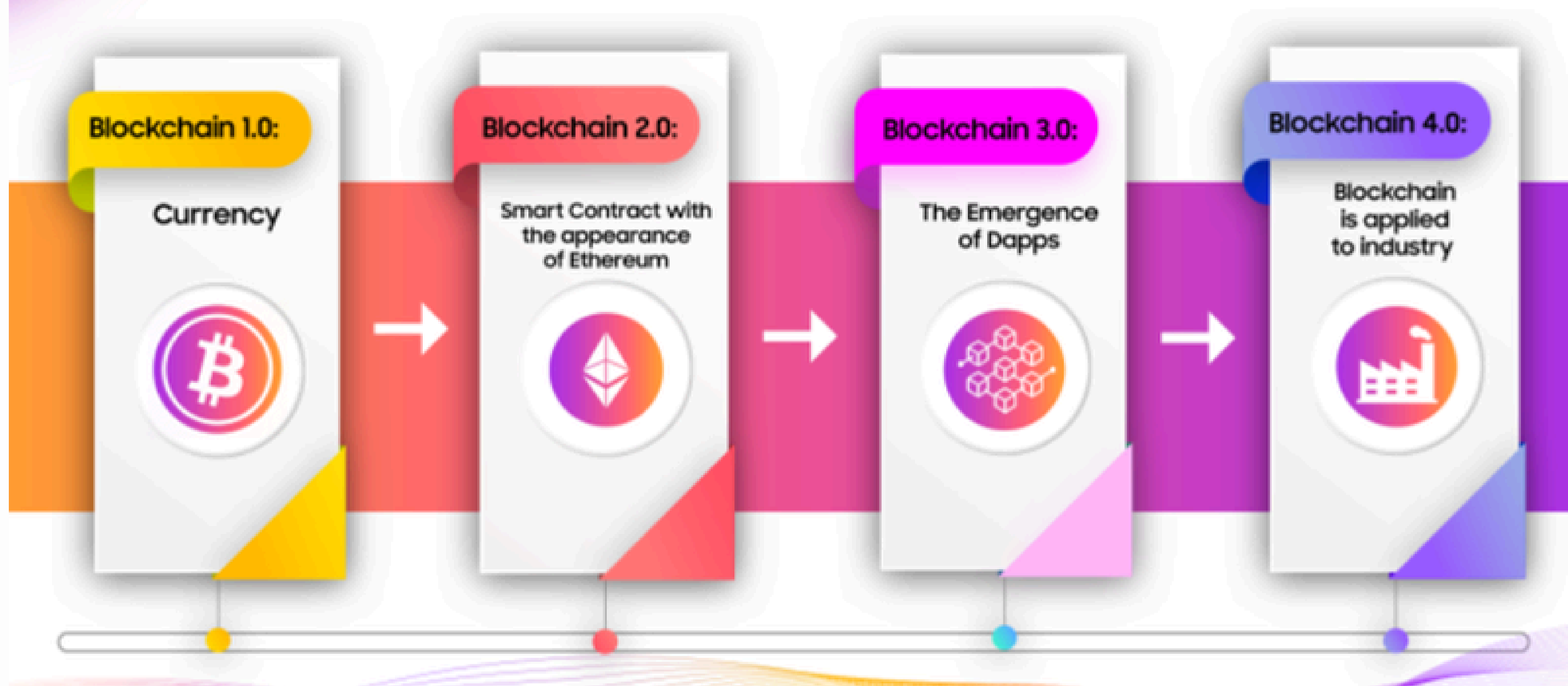
2 ID: 10b3-90fc

7/24/2023, 09:52:11

From bc1p-uq8m
To 2 Outputs

0.00015208 BTC • \$4.53
Fee 105.8K Sats • \$31.53

Blockchain Evolution



Blockchain 1.0

Blockchain 1.0:

Currency



Transactions

⬆️ Last **First** ⬆️ Value ⬆️ Value ⬆️ Fee ⬆️ Fee

0 ID: f31c-bb92 ⓘ 7/24/2023, 10:03:53		From Block Reward To 2 Outputs		6.45357181 BTC • \$192,311 Fee 0 Sats • \$0.00 ^
From		To		
1 Block Reward 0.00 BTC • \$0.00		1 3KZDwmJHB6QJ13QPXHhW7SS3yTESFPZoxb ⓘ ⓘ 6.45357181 BTC • \$192,311		
		2 Unknown 0.00000000 BTC • \$0.00		
TX 1 ID: d4d3-c640 ⓘ 7/24/2023, 09:52:07		From bc1p-ytjr ⓘ To 2 Outputs		0.00151686 BTC • \$45.20 Fee 94.5K Sats • \$28.17 ^
From		To		
1 bc1pk48s0mfwca8036tyq7vsdxwc54vyr54224hfn4c3grz50p2... ⓘ ⓘ 0.00246230 BTC • \$73.37		1 bc1pv6ar4n5ex7g0sf8h667xmv0g6k8d5j2zrlwgz343scnxkxpl0... ⓘ ⓘ ⬅️ 0.00121000 BTC • \$36.06		
		2 bc1pk48s0mfwca8036tyq7vsdxwc54vyr54224hfn4c3grz50p2sh... ⓘ ⓘ 0.00030686 BTC • \$9.14		
TX 2 ID: 10b3-90fc ⓘ 7/24/2023, 09:52:11		From bc1p-uq8m ⓘ To 2 Outputs		0.00015208 BTC • \$4.53 Fee 105.8K Sats • \$31.53 ^

NEXT BITCOIN HALVINGS

(years are estimated)

Genesis	-	Block reward 50
2012 Halving	-	Block reward 25
2016 Halving	-	Block reward 12.5
2020 Halving	-	Block reward 6.25
2024 Halving	-	Block reward 3.125
2028 Halving	-	Block reward 1.5625
2032 Halving	-	Block reward 0.78125
2036 Halving	-	Block reward 0.390625
2040 Halving	-	Block reward 0.1953125
2044 Halving	-	Block reward 0.09765625
2048 Halving	-	Block reward 0.04882812
2052 Halving	-	Block reward 0.02441406
2056 Halving	-	Block reward 0.01220703
2060 Halving	-	Block reward 0.00610351
2064 Halving	-	Block reward 0.00305175
2068 Halving	-	Block reward 0.00152587
2072 Halving	-	Block reward 0.00076293
2076 Halving	-	Block reward 0.00038146
2080 Halving	-	Block reward 0.00019073
2084 Halving	-	Block reward 0.00009536
2088 Halving	-	Block reward 0.00004768
2092 Halving	-	Block reward 0.00002384
2096 Halving	-	Block reward 0.00001192
2100 Halving	-	Block reward 0.00000596
2104 Halving	-	Block reward 0.00000298
2108 Halving	-	Block reward 0.00000149
2112 Halving	-	Block reward 0.00000074
2116 Halving	-	Block reward 0.00000037
2120 Halving	-	Block reward 0.00000018
2124 Halving	-	Block reward 0.00000009
2128 Halving	-	Block reward 0.00000004
2132 Halving	-	Block reward 0.00000002
2136 Halving	-	Block reward 0.00000001

WE ARE
HERE





What is a Bitcoin Halving? Explained by CoinGecko

CoinGecko

Bitcoin Halving



Watch on  YouTube



Bitcoin Halving Explained (Animated) - Why, How, and What it Means for You!

Whiteboard Crypto



Watch on  YouTube

